

Supervised learning -

SL is like teaching a computer how to do something by showing it examples. Hence models learn from labeled training data.

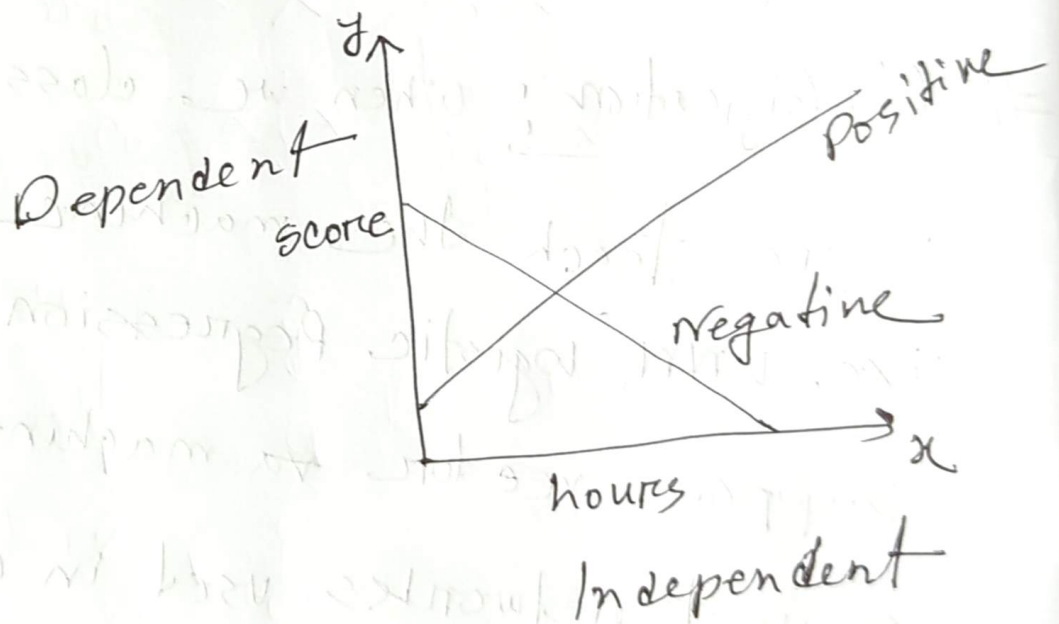
Classification: When we classify data before teach the machine. Algorithms like KNN, logistic Regression, Decision tree, Support vector machine and neural networks used in classification method.

Regression: Deal with numeric data. Algorithms like linear Regression, Polynomial Regr. and Regr. Trees used in Regr. method.

L-4

Regression —

Regr. is a statistical method that helps us to understand and predict the relationship between variables.



Linear Regression —

Equation: $y = mx + b$

Annotations:
- m : slope
- x : Independent variable
- y : value of y , $x=0$
- b : Dependent variable

* slope $\rightarrow$ how much y changes for a unit change in x .

Line Regression

Example

$$\begin{cases} * c = \bar{y} - (m\bar{x}) \\ * y = mx + c \\ * m = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} \end{cases}$$

Pizza size (x)	Pizza price (y)	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x}) * (y - \bar{y})$	$(x - \bar{x})^2$
8	10	-2	-3	6	4
10	13	0	0	0	0
12	16	2	3	6	4
		Total =		12	8

$$\bar{x} = \frac{8+10+12}{3} = 10$$

$$\bar{y} = \frac{13+10+16}{3} = 13$$

$$\therefore m = \frac{12}{8} = 1.5$$

$$\begin{aligned} c &= \bar{y} - (m\bar{x}) \\ &= 13 - (1.5 \times 10) \\ &= 13 - 15 = -2 \end{aligned}$$

$$\therefore y = 1.5x - 2 \quad (\text{Ans})$$

Gradient Descent -

$$(1) \quad \frac{\partial}{\partial c} J(m, c) = -\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)$$

$$(2) \quad \frac{\partial}{\partial m} J(m, c) = -\frac{1}{n} \sum_{i=1}^n x_i (y_i - \hat{y}_i)$$

$$(3) \quad m_1 = m_0 - \alpha \times \frac{\partial}{\partial m} J(m, c)$$

$$(4) \quad c_1 = c_0 - \alpha \times \frac{\partial}{\partial c} J(m, c)$$

Ex

x	y	$\hat{y}$
1	1	0
2	2	0
3	3	0
4	4	0

Let, $m=0, c=0$

$$\therefore \hat{y}_1 = 0 \times 1 + 0 = 0$$

$$\hat{y}_2 = 0 \times 2 + 0 = 0$$

We know,

$$\frac{\partial}{\partial m} J(m, c) = -\frac{1}{n} \sum_{i=1}^n x_i (y_i - \hat{y}_i)$$

$$= -\frac{1}{4} \langle 1 \times (1-0) + 2 \times (2-0) + 3 \times (3-0) + 4 \times (4-0) \rangle$$

$$= -\frac{1}{4} (1 + 4 + 9 + 16)$$

$$= -\frac{36}{4} = -7.5$$

$$\frac{\partial}{\partial c} J(m, c) = -\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)$$

$$= -\frac{1}{4} \langle (1-0) + (2-0) + (3-0) + (4-0) \rangle$$

$$= -2.5$$

$$m_1 = m_0 - \alpha \times \frac{\partial}{\partial m} J(m, c)$$

$$= 0 - 0.01 \times (-7.5) = 0.075$$

$$c_1 = c_0 - \alpha \times \frac{\partial}{\partial c} J(m, c)$$

$$= 0 - 0.01 \times (-2.5) = 0.025$$

- Compute new prediction -

$$\hat{y} = mx + c$$

$$\hat{y}_1 = 0.075 \times 1 + 0.025 = 0.1$$

$$\hat{y}_2 = 0.075 \times 2 + 0.025 = 0.175$$

$$\hat{y}_3 = 0.075 \times 3 + 0.025 = 0.25$$

$$\hat{y}_4 = 0.075 \times 4 + 0.025 = 0.325$$

Now, compute gradients -

$$\begin{aligned} \frac{\partial}{\partial m} J(m, c) &= -\frac{1}{4} \left\langle 1(1 - 0.1) + 2(2 - 0.175) \right. \\ &\quad \left. + 3(3 - 0.25) + 4(4 - 0.325) \right\rangle \\ &= -6.875 \end{aligned}$$

$$\begin{aligned} \frac{\partial}{\partial c} J(m, c) &= -\frac{1}{4} \left\langle (1 - 0.1) + (2 - 0.175) \right. \\ &\quad \left. + (3 - 0.25) + (4 - 0.325) \right\rangle \\ &= -2.2875 \end{aligned}$$

now, Update Parameters —

$$m_2 = m_1 - \alpha \times \frac{\partial}{\partial m} J(m, c)$$

$$= 0.075 - 0.01 \times (-6.875)$$

$$= 0.14375$$

$$c_2 = c_1 - \alpha \times \frac{\partial}{\partial c} J(m, c)$$

$$= 0.025 - 0.01 \times (-2.2875)$$

$$= 0.047875$$

(An.)

Logistic Regression

$$(1) h_{\theta}(x) = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x)}} \quad \left(\hat{y} = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x)}} \right)$$

$$(2) J(\theta) = -\frac{1}{n} \sum_{i=1}^n \left(y^{(i)} \log(h_{\theta}(x^{(i)})) + (1 - y^{(i)}) \log(1 - h_{\theta}(x^{(i)})) \right)$$

$$(3) \theta_j = \overset{\text{Zero}}{\theta_j} - \alpha \frac{\partial J(\theta)}{\partial \theta_j}$$

$$(4) \frac{\partial J(\theta)}{\partial \theta_0} = -\frac{1}{n} \sum_{i=1}^n (h_{\theta}(x^{(i)}) - y^{(i)})$$

$$(5) \frac{\partial J(\theta)}{\partial \theta_1} = -\frac{1}{n} \sum_{i=1}^n (h_{\theta}(x^{(i)}) - y^{(i)}) x^{(i)}$$

Ex

Hours(x)	Pass/Fail(y)	$\hat{y}(h_{\theta}(x^{(i)}))$
2	0	0.5
3	0	0.5
5	1	0.5
7	1	0.5

Now,

$$h_0(2) = \frac{1}{1 + e^{-(0 + 0 \times 2)}} = 0.5$$

$$h_0(3) = \frac{1}{1 + e^{-(0 + 0 \times 2)}} = 0.5$$

Cost

$$\therefore J(\theta) = -\frac{1}{n} \sum_{i=1}^n \left(y^{(i)} \log(h_0(x^{(i)})) + (1 - y^{(i)}) \log(1 - h_0(x^{(i)})) \right)$$

$$= -\frac{1}{4} \left[0 \log(0.5) + (1 - 0) \log(1 - 0.5) + 0 \log(0.5) + (1 - 0) \log(1 - 0.5) \right]$$

$$= -\frac{1}{4} \left[\log(0.5) + \log(0.5) + \log(0.5) + \log(0.5) \right]$$

$$= -\frac{1}{4} \times 4 \times (-0.693)$$

$$= 0.693$$

Now, compute Gradients —

$$\frac{\partial J(\theta)}{\partial \theta_0} = \frac{1}{n} \sum_{i=1}^n (\text{hnl}(x^{(i)}) - y^{(i)})$$

$$= \frac{1}{4} [(0.5 - 0) + (0.5 - 0) + (0.5 - 1) + (0.5 - 1)]$$

$$= \frac{1}{4} \times (-1) = -0.25$$

$$\text{and } \frac{\partial J(\theta)}{\partial \theta_1} = \frac{1}{n} \sum_{i=1}^n (\text{hnl}(x^{(i)}) - y^{(i)}) x^{(i)}$$

$$= \frac{1}{4} [(0.5 - 0) \times 2 + (0.5 - 0) \times 3$$

$$+ (0.5 - 1) \times 5 + (0.5 - 1) \times 7]$$

$$= \frac{1}{4} \times (-3.5) = -0.875$$

$$\therefore \theta_0 = \theta_0 - 0.1 \times (-0.25) \quad [\alpha = 0.1]$$

$$= 0 + 0.025 = 0.025$$

$$\theta_1 = \theta_1 - 0.1 \times (-0.875)$$

$$= 0 + 0.0875 = 0.0875$$

(Ans.)

Confusion Matrix —

	Positive	Negative
Positive	TP	FP
Negative	FN	TN

① Accuracy = $\frac{TP + TN}{TP + FP + FN + TN}$

② Precision = $\frac{TP}{TP + FP}$

③ Recall = $\frac{TP}{TP + FN}$

④ F1-score, $f_1 = 2 \times \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}$

Ex. A spam filter classify 10 emails,

$TP = 4, TN = 3, FP = 2, FN = 1$

$\therefore \text{Accuracy} = \frac{4+3}{4+3+2+1} \approx 0.7$

$\text{Precision} = \frac{4}{4+2} \approx 0.67$

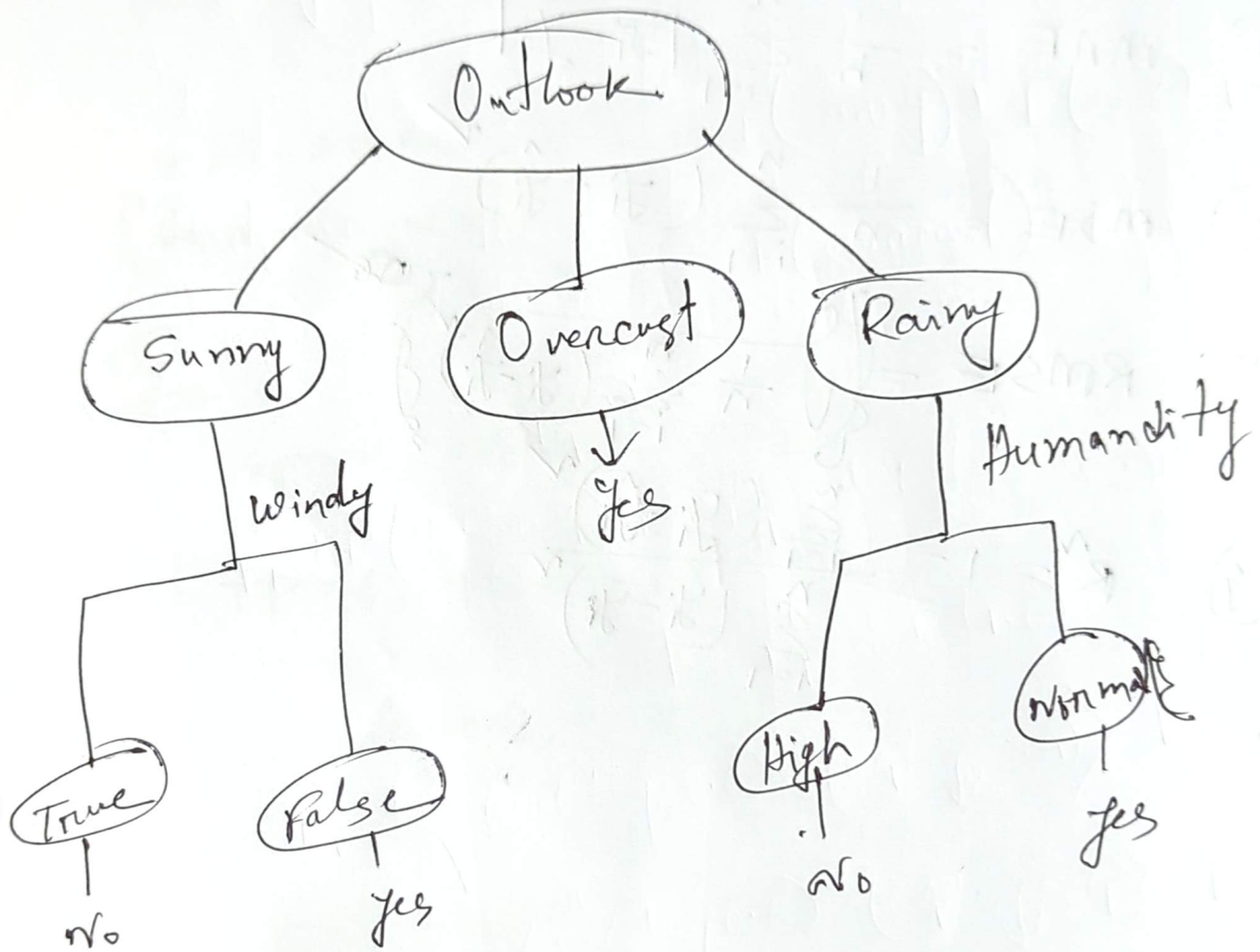
$$\text{Recall} = \frac{q}{q+1} = 0.8$$

$$\therefore f_1 = 2 \times \frac{0.67 \times 0.8}{0.67 + 0.8}$$

$$= 2 \times \frac{0.536}{1.47}$$

$$= 0.73 \text{ (Ans.)}$$

Decision Tree -



~~ID3~~ # Decision tree algorithm -

~~ID3~~

x(2) 34 C4.5

x(3) CART

* Entropy, $E(T) = - \sum_{i=1}^N P_i \log_2 P_i$

Play Golf	
Yes	No
9	5

$\therefore \text{Entropy}(\text{Play Golf}) = E(T, 9)$

$$= \left(\frac{9}{14} \log_2 \frac{9}{14} \right) + \left(\frac{5}{14} \log_2 \frac{5}{14} \right)$$

$$= 0.99$$

Target $\leftarrow$ $E(T, X)$ $\rightarrow$ Input

$$E(T, X) = \sum_{N \in X} P(N) E(N)$$

		Play Golf		Total	
		Yes	No		
Outlook	Rainy	2	3	5	5/14
	Sunny	3	2	5	5/14
	Overcast	4	0	4	4/14

$$E(\text{Play Golf}, \text{Outlook}) = P(\text{Rainy}) * E(2, 3) + \\ P(\text{Sunny}) * E(3, 2) + \\ P(\text{overcast}) * E(4, 0)$$

$$\Rightarrow \frac{5}{14} * E(0.6, 0.4) + \frac{5}{14} * E(0.6, 0.4) \\ + \frac{4}{14} * E(1, 0)$$

$$= \frac{5}{14} * \left\langle -(0.6 \log_2 0.6) - (0.4 \log_2 0.4) \right\rangle + \\ \frac{5}{14} * \left\langle -(0.6 \log_2 0.6) - (0.4 \log_2 0.4) \right\rangle + \\ \frac{4}{14} * \left\langle -(1 \log_2 1) - (0 \log_2 0) \right\rangle$$

$$= 0.693$$

$$\text{Information gain} = \text{Entropy}(T) - \text{Entropy}(T, x) \\ = 0.94 - 0.693 \\ = 0.247 \quad (\text{Ans})$$